Sameer Panghal

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EDUCATION

Georgia Institute of Technology, *Atlanta, GA* *Expected Graduation Date: Summer 2026*
M.S. Mechanical Engineering *(Concentration in Automation and Controls)*
Georgia Institute of Technology, *Atlanta, GA* *Aug 2022 – May 2025*
B.S. Mechanical Engineering *(Graduated with Highest Honors)*

WORK EXPERIENCE

Graduate Researcher – Advanced Crane Laboratory *Atlanta, GA* *May 2025 – Aug 2025*

- Developed a nonlinear Lagrangian dynamic model of payload motion in MATLAB, achieving 74% accuracy in simulation validation against real-world data.
- Implemented PID and Input Shaping algorithms to minimize payload sway during slewing/luffing; integrated real-time tip-over detection to actively prevent hazards.

Robotics Intern – Schaeffler Group (prev. Vitesco Technologies), *Seguin, TX* *May 2024 – Aug 2024*

- Designed and fabricated End-Of-Arm-Tooling (EOAT) in SOLIDWORKS (FDM/SLA) to handle 2.5kg+ loads with 6-DOF precision.
- Programmed Techman TM12S and UR10e cobots with Keyence vision sensors to automate rack loading; optimized placement logic to reduce cycle time, saving $720k/year.
- Deployed UR10e cobot scripts (Polyscript) to support ECU production lines for Ford Automotive.

PROJECT EXPERIENCE

Teleoperated Robotic Arm *Aug 2025 – Dec 2025*

- Architected a C/C++ control loop on STM32 (NUCLEO-L476RG) to drive ODrive S1 actuators via CANBUS, implementing Field-Oriented Control (FOC) with 16-bit encoder precision.
- Engineered 4-DOF custom actuators with 10:1 planetary gearboxes; validated torque requirements (18.8 Nm) via static load analysis (Safety Factor 1.3) and material selection.
- Designed a 600W power distribution PCB in KiCad using daisy-chained CAN topology to drive three 20A actuators and a 4th-axis gimbal.
- Built a Python/Jupyter pipeline to map vision model outputs directly to actuator setpoints for synchronized real-time teleoperation.

Tower Crane Vibration Reduction *Aug 2025 – Dec 2025*

- Engineered an adaptive Recursive Least Squares (RLS) algorithm on a linearized discrete-time model, reducing payload oscillations by ~85% in hardware validation.
- Tuned estimator forgetting factor $\lambda = 0.999$ to filter sensor noise, allowing accurate tracking of 20% changes in pendulum length during operation
- Implemented Zero Vibration Derivative (ZVD) shaping to counteract unreliable feedback (ζ approx. 0.4), ensuring suppression despite estimation errors17.

Variable Input Hand Controlled Ball Levitation *Jan 2025 – May 2025*

- Developed a PD-based height controller with Kalman filtering and moving average smoothing, maintaining ball position within ±2mm of target.
- Integrated ultrasonic sensors and brushless DC fans into a physics-based plant model, solving nonlinear airflow limitations through servo-actuated throttle control.

SKILLS

Embedded & Control: C/C++, Python, MATLAB/Simulink, STM32, CANBUS, I2C, FOC, ROS, Git, Raspberry Pi
Mechanical Design: SOLIDWORKS, CATIA, Onshape, GD&T, DFM/DFA, Motor Sizing, KiCad (PCB)
Fabrication: 3D Printing (FDM, SLA, SLS), CNC Machining, Laser Cutting